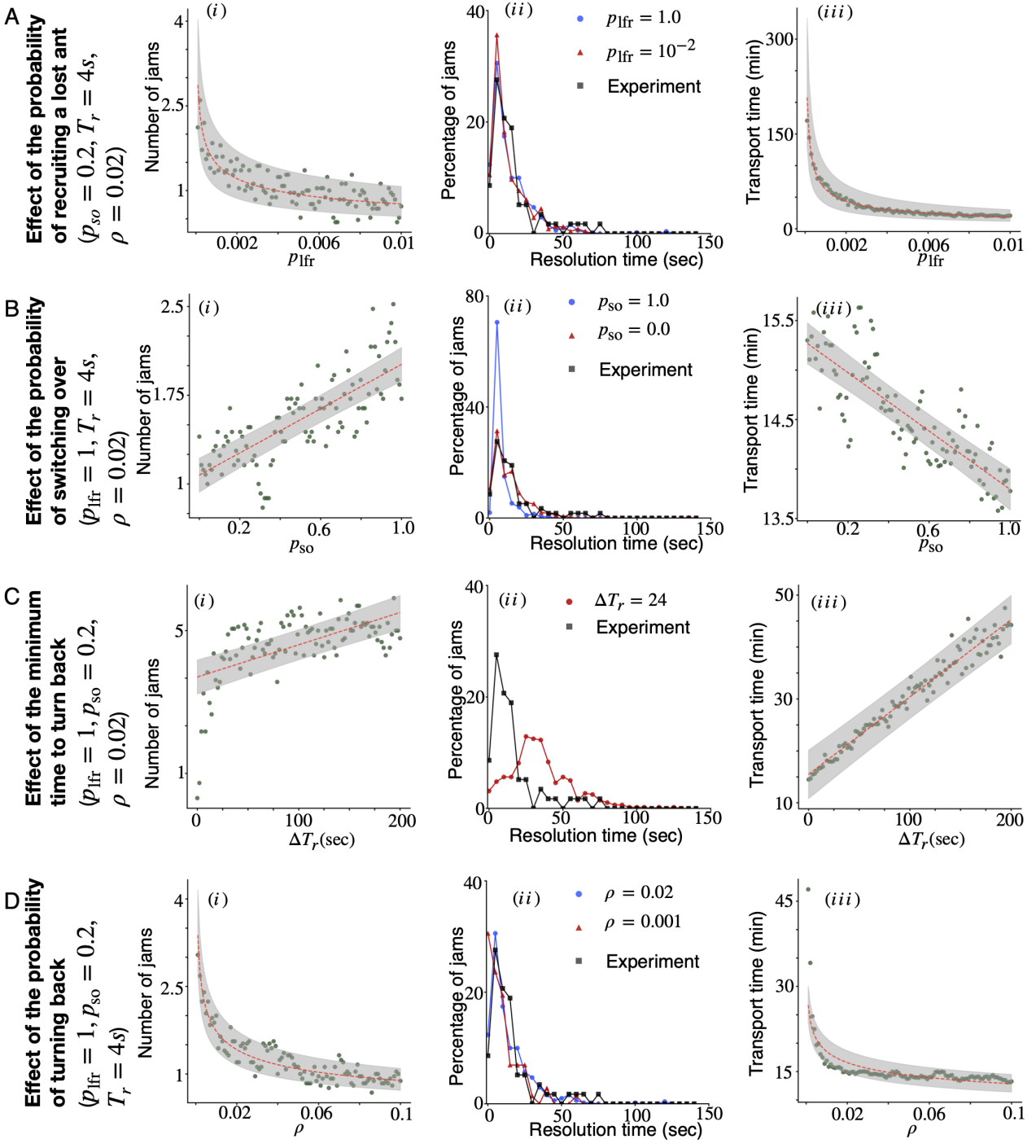

A
Effect of the probability of recruiting a lost ant ($p_{so} = 0.2, T_r = 4s, \rho = 0.02$)
(i)
Number of jams
p_{lfr}
(ii)
Percentage of jams
Resolution time (sec)
$p_{\mathrm{lfr}} = 1.0$
$p_{\mathrm{lfr}} = 10^{-2}$
Experiment
(iii)
Transport time (min)
p_{lfr}
B
Effect of the probability of switching over ($p_{\mathrm{lfr}} = 1, T_r = 4s, \rho = 0.02$)
(i)
Number of jams
p_{so}
(ii)
Percentage of jams
Resolution time (sec)
$p_{\mathrm{so}} = 1.0$
$p_{\mathrm{so}} = 0.0$
Experiment
(iii)
Transport time (min)
p_{so}
C
Effect of the minimum time to turn back ($p_{\mathrm{lfr}} = 1, p_{\mathrm{so}} = 0.2, \rho = 0.02$)
(i)
Number of jams
ΔT_r(sec)
(ii)
Percentage of jams
Resolution time (sec)
$\Delta T_r = 24$
Experiment
(iii)
Transport time (min)
ΔT_r(sec)
D
Effect of the probability of turning back ($p_{\mathrm{lfr}} = 1, p_{\mathrm{so}} = 0.2, T_r = 4s$)
(i)
Number of jams
ρ
(ii)
Percentage of jams
Resolution time (sec)
$\rho = 0.02$
$\rho = 0.001$
Experiment
(iii)
Transport time (min)
ρ